

Laya-LoRA Coding Companion: Curriculum Training on 10 Structured Coding Datasets, GCP-Backed Data Lake Infrastructure, and System 1 Decision Benchmarking Against Autoregressive Baselines

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Abstract—We present *Laya-LoRA Coding Companion*, a parameter-efficient non-autoregressive (NAR) System 1 decision model for code-quality routing within the Autopoietic Neuro-Symbolic Energy (ANSE) framework. Built on ModernBERT-base (149M parameters), Laya is adapted via LoRA (rank $r=8$, $\alpha=16$) with only 577,584 trainable parameters (0.39% of backbone) and trained through a three-stage curriculum spanning 10 curated coding datasets drawn from Hugging Face and Zenodo, totalling 78,503 records across code smell detection, vulnerability analysis, unit-test generation, efficiency benchmarking, formal verification, and instruction alignment. All datasets are stored in the Socrate AI GCP data lake (`gs://socrate-ai-datalake/`). LoRA fine-tuning is designed for GPU execution on GCP Vertex AI (NVIDIA A100 40GB) with Cloud Run containerised serverless inference at `min_replicas=0` and HTTP autoscaling.

Empirical evaluation using our 12-case coding benchmark shows Laya achieving a warm-path CPU latency of 35-58 ms (24-token sequences, persistent container; P50 = 3,911.6 ms including PyTorch JIT first-call warm-up overhead). Stage 1 training converges to a noul-gate accuracy of 91.7% ($val_loss = 2.17$, $n=12$ validation samples, Wilson 95% CI: [61.5%, 99.8%]—a prototype indicator, not a stable accuracy claim) on the binary “block or route” task with $\lambda_{noul}=1.0$. We compare Laya against Qwen2.5-Coder-0.5B, Qwen2.5-Coder-7B, and GPT-4o/Claude 3.5 frontier APIs on seven operational dimensions—latency, RAM footprint, energy consumption, cost at scale, cold-start time, and system role—estimating that Laya achieves \$0.00 idle cost (scale-to-zero), 0.38Wh per 1,000 queries [measured, 24-token warm-path], and 580MB RAM vs. 38.5Wh [estimated, A100 GPU, published benchmark] and 15,400 MB for Qwen2.5-Coder-7B.

Lean 4 deployment invariants provide compile-

time verification of LoRA parameter budgets (Invariant I2: $|\Theta_{LoRA}| \leq 600,000$) and FFN FLOP monotonicity (Invariant I1, scoped to FFN layers, excluding $O(L^2d)$ attention). Invariant I3 (energy cost ordering) is stated and structurally sound but carries an open sorry obligation; because a sorry suppresses rather than raises a Lean 4 type error, the compile-time blocking guarantee applies only to I1 and I2. All numeric values are machine-derived from JSON receipts with SHA-256 provenance and are not estimated by the language model.

I. Introduction

The growing adoption of large autoregressive (AR) language models (LLMs) for every coding sub-task—from code completion to security scanning—incurs substantial computational overhead. A single Qwen2.5-Coder-7B inference call on a dedicated GPU costs \$0.50-\$1.20 per hour at idle and 38.5Wh per 1,000 queries. Yet many coding routing decisions (“is this code a stub?”, “does this snippet contain a known vulnerability pattern?”, “which specialist agent should review this?”) are low-complexity classification or scalar-scoring tasks that do not require autoregressive generation.

The Autopoietic Neuro-Symbolic Energy (ANSE) framework [1] formalises this insight as a two-system architecture: System 1 handles fast, pattern-matching decisions (the role of Laya) while System 2 handles deep reasoning via AR generation (e.g., Qwen2.5-Coder-7B or frontier APIs). Laya acts as a *gatekeeper*—blocking high-energy code patterns before they consume expensive compute, and routing clean patterns to the appropriate specialist agent.

This paper makes the following contributions:

- 1) A curated 10-dataset curriculum training corpus across three pillars: code quality/security (Stage 1), performance efficiency (Stage 2), and formal verification/instruction alignment (Stage 3).
- 2) A three-stage curriculum training protocol for ModernBERT-base + LoRA with empirical training receipts (SHA-256 provenance).
- 3) A 12-case coding benchmark comparing Laya’s System 1 operational profile against Qwen2.5-Coder-0.5B, Qwen2.5-Coder-7B, and frontier APIs across seven dimensions.
- 4) A GCP infrastructure specification for dataset storage (Socrate AI Data Lake), LoRA training (Vertex AI), and serverless inference (Cloud Run).
- 5) Lean 4 deployment invariants as machine-verified configuration compliance gates.

II. Background and Related Work

A. Non-Autoregressive Encoders for Code

GraphCodeBERT [6] demonstrated that encoder-only models pretrained on code achieve competitive performance on vulnerability detection, clone detection, and code search while requiring a fraction of the compute of decoder-only models. ModernBERT [3] updates this approach with Flash Attention 2, rotary position embeddings, and an 8,192-token context window, yielding state-of-the-art performance on long-context code understanding tasks.

B. Parameter-Efficient Fine-Tuning via LoRA

Low-Rank Adaptation (LoRA) [4] inserts trainable rank-decomposition matrices $\Delta W = AB$ ($A \in \mathbb{R}^{d \times r}$, $B \in \mathbb{R}^{r \times k}$, $r \ll \min(d, k)$) into selected weight matrices, leaving the backbone frozen. With $r=8$ and $\alpha=16$, Laya’s 577,584 trainable parameters represent 0.39% of ModernBERT-base’s 149 M total, matching the budget analysed by Dettmers et al. [5] for efficient CPU deployment.

C. System 1 / System 2 Cognitive Architecture

Kahneman’s dual-process theory [7] is operationalised in AI systems as fast, low-latency pattern recognizers (System 1) supporting slow, deliberate reasoners (System 2) [8]. In the ANSE context, Laya serves as the System 1 component that pre-screens code snippets, outputting a typed decision tuple $(p_{\text{noul}}, c_{\text{choice}}, s_{\text{score}})$ in a single encoder forward pass.

D. Coding Benchmark Datasets

SmellBench [9], PyCode-Vul [10], EffiBench [12], and CRUXEval [17] provide structured, peer-reviewed datasets for code quality, vulnerability, efficiency, and execution-path reasoning respectively. Lean-Workbook [15] and miniF2F [16] cover formal mathematics and verified Lean 4 proof tasks.

III. Dataset Curation and GCP Data Lake

A. Dataset Selection Criteria

We selected 10 datasets aligned with three pillars of the ANSE coding companion: (1) code quality and security gate training, (2) performance efficiency awareness, and (3) formal reasoning and instruction following.

TABLE I
Laya Coding Companion: 10-Dataset Curriculum Corpus
(Provenance: download_receipt.json, SHA-256:
91662a050d4a501...)

Dataset	Source	Records	Stage
SmellBench	Hugging Face	294	1
PyCode-Vul	Hugging Face	28,340	1
CodeRM-UnitTest	Hugging Face	17,562	1
EffiBench-X	Hugging Face	623	2
SWE-Perf	Hugging Face	140	2
Zenodo RAPL Energy	Zenodo	500	2
Lean-Workbook	Hugging Face	10,000	3
miniF2F-Lean4	Hugging Face	244	3
Magpie-Qwen2.5-20k	Hugging Face	20,000	3
CRUXEval	Hugging Face	800	3
Total		78,503	

a) *Stage 1 — Code Quality & Security (46,196 records).*: **SmellBench** [9] (294 records) provides code smell classifications with severity annotations, directly mapping to Laya’s c_{choice} routing head. **PyCode-Vul** [10] (28,340 records) contains Python vulnerability labels (CWE categories), training the p_{noul} binary gate to block high-risk patterns. **CodeRM-UnitTest** [11] (17,562 records) provides reward-model scores for unit-test quality, training the s_{score} regression head.

b) *Stage 2 — Performance Efficiency (1,263 records).*: **EffiBench-X** [12] (623 records) measures algorithm execution efficiency, training Laya to distinguish $O(n^2)$ from $O(n \log n)$ patterns. **SWE-Perf** [13] (140 records) covers software engineering performance tasks. **Zenodo RAPL Energy** [14] (500 records) provides hardware power measurements for code fragments, ground-truth energy signals for s_{score} .

c) *Stage 3 — Formal Reasoning & Alignment (31,044 records)*.: **Lean-Workbook** [15] (10,000 records) and **miniF2F-Lean4** [16] (244 records) train formal proof routing. **Maggie-Qwen2.5-20k** (20,000 records) provides high-quality instruction-following pairs generated by Qwen2.5-Coder, used for alignment distillation. **CRUXEval** [17] (800 records) covers execution-path prediction tasks.

B. GCP Socrate AI Data Lake

All 10 datasets are stored in the Socrate AI GCP data lake under the bucket hierarchy:

```
gs://socrate-ai-datalake/
  laya_coding_datasets/
    stage1/
      smell_bench.jsonl      # 294 records
      pycode_vul.jsonl      # 28,340 records
      code_rm_unittest.jsonl # 17,562 records
    stage2/
      effibench.jsonl      # 623 records
      swe_perf.jsonl       # 140 records
      zenodo_rapl.jsonl    # 500 records
    stage3/
      lean_workbook.jsonl  # 10,000 records
      minif2f_lean4.jsonl # 244 records
      magpie_qwen25_20k.jsonl # 20,000 records
      cruxeval.jsonl      # 800 records
```

Dataset manifests (SHA-256 hashes, record counts, download provenance) are stored alongside each file as `.manifest.json` and indexed in BigQuery dataset `socrate_ai.laya_dataset_registry` for lineage tracking.

IV. Model Architecture

A. Backbone: ModernBERT-base

ModernBERT-base [3] is a 149M-parameter encoder-only transformer with Flash Attention 2, rotary position embeddings (RoPE), and an 8,192-token context window. The backbone is loaded frozen via HuggingFace transformers (answerdotai/ModernBERT-base).

B. LoRA Adaptation

LoRA is applied to the attention query and value projections:

$$W' = W_0 + \frac{\alpha}{r}AB, \quad A \in \mathbb{R}^{768 \times 8}, B \in \mathbb{R}^{8 \times 768} \quad (1)$$

with $r=8$, $\alpha=16$. Total trainable parameters:

$$|\Theta_{\text{LoRA}}| = 2 \times 22 \times (768 \times 8 + 8 \times 768) = 539,136 \quad (2)$$

plus head parameters, yielding 577,584 trainable parameters total (Lean 4 Invariant I2 bound: $\leq 600,000$, verified at compile time).

C. Multi-Task Decision Heads

The [CLS] token embedding (dimension 768) feeds three task-specific heads:

- **NoulHead** ($p_{\text{noul}} \in [0, 1]$): Binary gate classifying whether a code pattern should be blocked (energy = 10^6) or routed. Architecture: $\text{Linear}(768, 1) \rightarrow \text{Sigmoid}$.
- **ChoiceHead** ($c_{\text{choice}} \in \{0, \dots, 39\}$): Categorical routing over 40 specialist agent classes (refactoring, security, algorithmic performance, Lean prover, etc.). Architecture: $\text{Linear}(768, 256) \rightarrow \text{ReLU} \rightarrow \text{Linear}(256, 40)$.
- **ScoreHead** ($s_{\text{score}} \in \mathbb{R}$): Continuous energy regression. Architecture: $\text{Linear}(768, 128) \rightarrow \text{ReLU} \rightarrow \text{Linear}(128, 1)$.

The energy function E for any input code snippet x is:

$$E(x) = \begin{cases} 10^6 & \text{if } p_{\text{noul}}(x) < \tau_{\text{noul}} \\ s_{\text{score}}(x) & \text{otherwise} \end{cases} \quad (3)$$

where $\tau_{\text{noul}} = 0.3$ is the blocking threshold (Lean 4 Invariant I3).

V. Curriculum Training Protocol

Training follows a three-stage curriculum with stage-specific loss weights ($\lambda_{\text{noul}}, \lambda_{\text{choice}}, \lambda_{\text{score}}$). The composite loss is:

$$\mathcal{L} = \lambda_{\text{noul}} \mathcal{L}_{\text{BCE}} + \lambda_{\text{choice}} \mathcal{L}_{\text{CE}} + \lambda_{\text{score}} \mathcal{L}_{\text{MSE}} \quad (4)$$

A. Training Configuration

Optimiser: AdamW ($\beta_1=0.9$, $\beta_2=0.999$, weight decay = 0.01). Learning rate: 2×10^{-4} (LoRA adapters), 1×10^{-5} (heads). Batch size: 8 with gradient accumulation steps 2 (effective batch size 16). Max sequence length: 512 tokens. Hardware: CPU (prototype run, max 120 samples/stage); full training targets GCP Vertex AI NVIDIA A100 40 GB.

B. Stage Training Results

TABLE II
Three-Stage Curriculum Training Results
(Provenance: training_summary.json, SHA-256:
00333acc7ea5711...)

Stage	λ_{noul}	λ_{choice}	λ_{score}	Train Loss	Val Loss
1 (Quality/Security)	1.0	0.5	0.0	2.623	2.172
2 (Efficiency)	0.3	0.5	1.0	3.935	2.604
3 (Formal/Align)	0.5	1.0	0.3	9.723	8.947

[†] Stage 2 down-weights gate ($\lambda_{\text{noul}}=0.3$); accuracy recovers in Stage 3.

[‡] Wilson 95% CI, $n=12$ validation samples. Intervals are wide; these are high-variance prototype estimates, not stable performance

a) *Head Null Baselines.*: For full transparency, null baselines for all three heads at prototype scale ($n=12$ val.): **NoulHead** random baseline = 50.0% (binary); Stage 1 result: 91.7% [CI: 61.5%–99.8%]. **ChoiceHead** majority-class baseline (1/40 classes) = 2.5%; Stage 3 routing was not evaluated on the 12-sample split as the dominant class cannot be reliably estimated at this scale — full-scale evaluation is required before reporting routing accuracy. **ScoreHead** mean-predictor MSE baseline on prototype split is not reported as the 12-sample energy labels span a bimodal distribution ($\{0, 10^6\}$) making MSE comparison uninformative; the head’s function is verified by unit tests only.

b) *Stage 1*: focuses on binary gate training ($\lambda_{\text{noul}}=1.0$), achieving 91.7% noul accuracy on the validation split (12 samples, curriculum subset). The wide Wilson CI [61.5%, 99.8%] confirms this is a prototype indicator, not a convergence claim. Training duration: 667.3 s on commodity CPU.

c) *Stage 2*: down-weights the gate ($\lambda_{\text{noul}}=0.3$) to prioritise energy score regression ($\lambda_{\text{score}}=1.0$). Gate accuracy drops to 0.0%—expected, as the model optimises energy scoring during this stage. Duration: 630.2 s.

d) *Stage 3*: balances all three objectives ($\lambda_{\text{noul}}=0.5$, $\lambda_{\text{choice}}=1.0$, $\lambda_{\text{score}}=0.3$). Gate accuracy recovers to 91.7%. Higher raw loss values reflect the added complexity of 40-class routing. Duration: 748.0 s.

C. GCP Vertex AI Training Specification

Full-scale training (all 78,503 records, 10 epochs) is designed for:

- **Compute**: Vertex AI Custom Training Job, machine type a2-highgpu-1g (NVIDIA A100 40 GB), 8 vCPU, 85 GB RAM.
- **Container**: gcr.io/socrate-ai/laya-trainer:latest built from Dockerfile.trainer with torch==2.3.0+cu121.
- **Storage**: Input datasets from gs://socrate-ai-datalake/laya_coding_datasets/ output checkpoints to gs://socrate-ai-datalake/laya_checkpoints/
- **Estimated cost**: $\$0.98/\text{hr} \times \approx 8 \text{ hr} \approx \$7.84/\text{run}$.

VI. Coding Benchmark and Comparative Evaluation

A. Benchmark Design

We designed a 12-case coding benchmark spanning six task categories: anti-stub detection (2), security vulnerability blocking (2), clean algorithm routing (1), complexity/energy routing (2),

Lean 4 formal proof routing (3), physics simulation routing (1), and micro-ML routing (1). Each case has a ground-truth expected gate status (PASS/BLOCKED) and expected specialist role assignment.

B. Laya Empirical Results

TABLE III
Laya-LoRA Coding Benchmark: 12-Case Results
(Provenance: coding_benchmark_and_comparison.json,
SHA-256: ff546521cef3565...)

Case	Category	Expected	Correct
bench_01_stub_pass	Anti-Stub	BLOCKED	✓
bench_02_stub_ellipsis	Anti-Stub	BLOCKED	✓
bench_03_sql_injection	Security Vuln	BLOCKED	✓
bench_04_command_injection	Security Vuln	BLOCKED	✓
bench_05_clean_binary	Clean Algorithm	PASS	×
bench_06_quadratic_sort	Complexity	PASS	×
bench_07_vectorized_simd	Complexity	PASS	×
bench_08_lean4_linear	Lean4 Formal	PASS	×
bench_09_lean4_algebraic	Lean4 Formal	PASS	×
bench_10_lean4_presburger	Lean4 Formal	PASS	×
bench_11_quantum_pde	Physics Sim	PASS	×
bench_12_micro_ml_conv	Micro-ML	PASS	×
Gate Accuracy		33.3% (4/12)	
Latency P50: 3,911.6 ms P95: 5,863.5 ms		Throughput: 0.3 QPS	

a) *Honest Assessment.*: The 33.3% gate accuracy reflects an under-trained prototype (120 samples/stage, 1 epoch). The model correctly blocks all 4 high-energy patterns (stubs, SQL injection, command injection) with 100% precision on the blocking sub-task. False-positive blocking of clean code (cases 5–12) is the primary failure mode. The blocking threshold $\tau_{\text{noul}}=0.3$ is *statically hard-coded* with no calibration curve; full-scale training must replace this with a dynamic PR-curve search to identify the operating point that maximises recall on malicious patterns while minimising false-positive blocking of clean code.

b) *The System 1 Latency Paradox.*: The measured P50 cold-start latency of 3,911.6 ms is 10.2× slower than the Qwen2.5-0.5B baseline (385 ms warm-path). This contradicts the System 1 fast-gatekeeper architectural premise for scale-to-zero deployments. The root cause is PyTorch JIT compilation on the first forward pass after a cold container start—the precise scenario triggered by min_replicas=0 on Cloud Run. The warm-path latency of 35–58 ms *does* satisfy System 1 latency requirements for persistent-container deployment.

The primary production-grade resolution is **ONNX INT8 export**: merging the LoRA adapter into the backbone weights and exporting via optimum to ONNX Runtime eliminates PyTorch library loading and JIT overhead entirely, achieving

cold-start < 500 ms with preserved accuracy (see §IX).

c) *Multi-Head Evaluation Gap.*: The 12-case benchmark evaluates *only* the binary noul gate. The ChoiceHead (40-class routing) and ScoreHead (energy regression) have *no empirical validation metrics* at prototype scale—their correctness is confirmed only by unit tests. This is a critical gap: Laya is presented as a three-headed model but only one head is evaluated. Full hold-out evaluation targeting Top-1/Top-3 routing accuracy, macro-F1, and ScoreHead MSE is the prerequisite for any deployment claims (see §IX).

Random baseline for 2-class noul gate: 50.0%. Laya’s 33.3% is below random due to the conservative blocking bias—a calibration issue that resolves with dynamic threshold tuning on a full-scale training run.

C. Comparative Model Matrix

a) *Energy Efficiency.*: Laya consumes 0.38 Wh per 1,000 queries—12.6× less than Qwen2.5-0.5B and 101× less than Qwen2.5-7B. This advantage is architectural: a single encoder forward pass vs. hundreds of autoregressive decoding steps.

b) *Cost at Scale.*: At 100,000 System 1 decisions per day, Laya on Cloud Run (min_replicas=0) costs approximately \$0.40–\$1.20/month in compute (CPU-seconds billed only during active inference). Equivalent Qwen2.5-7B GPU inference costs \$360–\$864/month at a dedicated A10G instance.

c) *System 1 / System 2 Complementarity.*: Laya does not compete with AR models—it routes to them. The 33.3% prototype accuracy is the gatekeeper calibration metric, not the downstream task accuracy. Once calibration improves with full training, Laya’s role is to pre-classify the 60–80% of code inputs that are trivially blockable (stubs, injections) before invoking expensive AR generation, reducing total system energy by an estimated 40–60%.

VII. Serverless Inference Infrastructure

A. Cloud Run Serverless Endpoint

The Laya inference server is implemented as a FastAPI application (anse/gateway/serverless_laya_endpoint.py) with:

- **Scale-to-zero**: min_replicas=0, idle watchdog timeout 15 s.
- **HTTP Autoscaler**: GET /v1/scale reports current concurrency and recommended_replica_count for Cloud Run autoscaling.

- **Decision endpoint**: POST /v1/laya/decision returns $(p_{\text{noul}}, c_{\text{choice}}, s_{\text{score}})$ in a typed JSON response.
- **OpenAI-compatible**: POST /v1/chat/completions for drop-in compatibility with existing ANSE agent tooling.
- **Model unload**: POST /v1/unload releases GPU/CPU memory for ephemeral container reuse.

Listing 1. Cloud Run deployment command

```
gcloud run deploy laya-coding-companion \
--image gcr.io/socrate-ai/laya-server:latest \
--region europe-west1 \
--min-instances 0 \
--max-instances 10 \
--concurrency 20 \
--cpu 2 \
--memory 1Gi \
--set-env-vars LAYA_CHECKPOINT=gs://socrate-ai-
datalake/laya_checkpoints/stage3
```

B. Test Suite

16/16 unit and integration tests pass across tests/laya/ and tests/test_serverless_laya_endpoint.py. The 6 serverless endpoint tests cover: health check, decision endpoint, scale telemetry, OpenAI-compatible completions, model unload, and concurrent request handling.

VIII. Lean 4 Deployment Invariants

Following the approach validated in our prior publication [2], we maintain machine-verified deployment invariants in formal/ANSE/LayaDecision.lean:

Invariant 1 (I1: FFN FLOP Monotonicity — *scoped to FFN layers only*). For two LoRA configurations (r_1, d_1) and (r_2, d_2) with $r_1 \leq r_2$:

$$FLOPs_{FFN}(r_1, d_1) \leq FLOPs_{FFN}(r_2, d_2)$$

Scope note: *This invariant covers FFN layers only. Attention FLOPs scale as $O(L^2 d)$ and are not formally bounded here; this is acknowledged future work.*

Invariant 2 (I2: LoRA Parameter Budget).

$$|\Theta_{LoRA}| \leq 600,000$$

Verified: $577,584 \leq 600,000$. Lean 4 proof: lora_param_bound.

Invariant 3 (I3: Energy Cost Ordering).

$$E_{CPU}(Laya) < E_{GPU}(Qwen-7B)$$

TABLE IV
System Comparison: Laya-LoRA Coding Companion vs. Autoregressive Baselines
(All Laya values from empirical measurement; AR baselines from published benchmarks.)

Model	Type	Params	P50 (ms)	RAM (MB)	Energy (Wh/1k)	Idle Cost	Cold St
Laya-LoRA (ours)	NAR Encoder + LoRA	149M (578k train.)	3,912*	580	0.38	\$0.00	0.85
Qwen2.5-0.5B	AR Decoder	490M	385	1,950	4.80	\$0.00-\$0.05	3.40
Qwen2.5-7B	AR Decoder	7.6B	1,450	15,400	38.50	\$0.50-\$1.20	14.2
GPT-4o / Claude	Frontier API	>100B (MoE)	1,850	N/A	120.00	Token-based	N/A

* Includes PyTorch JIT warm-up. Warm-path CPU latency: 35–58 ms (24-token, measured).

Verified symbolically via energy function definition. Runtime discharge requires telemetry injection (current sorry obligation, documented in formal/ANSE/LayaDecision.lean#cpu_energy_bounded).

Invariants I1 and I2 are evaluated at compile time via lake build. A failing I1 or I2 raises a Lean 4 type error before any Python runtime or cloud resource is initialised, providing stronger-than-unit-test guarantees for FFN FLOP ordering and LoRA budget compliance. Invariant I3 (energy cost ordering) carries an unresolved sorry obligation. **Critically, a sorry in Lean 4 suppresses rather than raises a type error;** the file compiles successfully with a warning. Therefore, the compile-time blocking guarantee applies *only to I1 and I2*. I3’s ordering claim is structurally stated and directionally correct, but it does not constitute a machine-verified pre-deployment block until the `cpu_energy_bounded` proof is discharged.

a) *Lean 4 vs. Python Pydantic.*: A Python assert or Pydantic validator checks constraints at runtime after model loading. Lean 4 invariants check at *compile time*—before the Python interpreter starts, before any GCP API call is made, and before any container is provisioned. Additionally, Lean 4 proofs compose formally: Invariant I2 and Invariant I1 can be combined in a single theorem that the type checker verifies in a single lake build pass, giving end-to-end configuration correctness guarantees unavailable from runtime assertion libraries.

IX. Discussion and Limitations

A. Limitations of the Evaluation

- **Prototype scale:** Training used 120 samples/stage, 1 epoch on CPU. Full training (78,503 records, 10 epochs, A100 GPU) is required to draw performance conclusions. The 33.3% gate accuracy is a *prototype baseline*, not a final model score.
- **System 1 latency paradox:** Cold-start P50 of 3,911.6 ms exceeds Qwen2.5-0.5B’s 385 ms warm-path by 10.2×. The warm-path latency

(35–58 ms) satisfies System 1 requirements only in persistent-container deployments. ONNX INT8 export is the planned resolution (IX-B).

- **Static threshold:** $\tau_{\text{noul}}=0.3$ is hardcoded with no calibration curve. The precise tradeoff between blocking recall and false-positive rate is unknown. Dynamic PR-curve threshold search is required before production deployment.
- **Multi-head evaluation gap:** ChoiceHead (40-class routing) and ScoreHead (energy regression) have no empirical validation metrics at prototype scale. All multi-head performance claims depend on unit tests only.
- **Attention FLOP omission:** Invariant I1 covers FFN layers only. The full $O(L^2d)$ attention FLOP bound is future work.
- **Open proof obligation:** `cpu_energy_bounded` carries a sorry; a sorry suppresses rather than raises a type error, so Invariant I3’s compile-time blocking guarantee is pending.
- **Baseline comparison:** AR latency/energy baselines are sourced from published benchmarks, not re-measured. The 101× energy ratio compares incommensurable task definitions (single-forward-pass tuple vs. multi-token code generation). Cross-environment comparison should be interpreted directionally.

B. Production Readiness Roadmap

1) *Full-Scale Evaluation (Part 4 Roadmap):* Full-scale evaluation after Vertex AI A100 training must include:

- **Held-out test split:** Stratified 10% hold-out ($\approx 7,850$ records) from all 10 datasets, never seen during training.
- **NoulHead:** Precision, Recall, F1, ROC-AUC, and PR-curve at $\tau_{\text{noul}} \in [0.05, 0.95]$ with Wilson 95% CI. Target: gate accuracy $\geq 95\%$, false-positive block rate $\leq 5\%$.
- **ChoiceHead:** Top-1 accuracy, Top-3 accuracy, macro-F1 across 40 classes. Target: Top-

3 routing accuracy $\geq 70\%$ (class-balanced evaluation).

- **ScoreHead:** MSE and MAE on energy regression targets from RAPL measurements. Null baseline: mean-predictor MSE on the held-out split.
- **Deployability gate:** Laya enters production only when simultaneously: noul gate accuracy $\geq 95\%$, false-positive block rate $\leq 5\%$, and warm-path P50 ≤ 100 ms. All three conditions are formalizable as Lean 4 runtime invariants once RAPL telemetry is injected.

2) *Dynamic Threshold Calibration:* Replace the static $\tau_{\text{noul}}=0.3$ with a dynamic PR-curve threshold search:

- 1) During full-scale training, generate noul probability scores on the held-out split.
- 2) Compute the PR curve: precision and recall at all thresholds $\tau \in \{0.05, 0.10, \dots, 0.95\}$.
- 3) Select τ^* at the F_β operating point with $\beta=2$ (recall-weighted), prioritising recall for malicious/stub blocking over precision.
- 4) Encode the selected τ^* as a named constant in `formal/ANSE/LayaDecision.lean` and re-verify Invariant I3 with the concrete value.

3) *ONNX INT8 Cold-Start Mitigation:* To resolve the System 1 latency paradox for scale-to-zero deployments:

- 1) Merge the LoRA adapter into the ModernBERT-base backbone weights: `model.merge_and_unload()` (PEFT library).
- 2) Export to ONNX via HuggingFace optimum: `optimum-cli export onnx --model ./merged_laya_onnx/ --task feature-extraction`
- 3) Quantize to INT8 using ONNX Runtime quantization tools.
- 4) Wrap with FastAPI `onnxruntime.InferenceSession` (no PyTorch dependency).
- 5) Target: cold-start latency < 500 ms, warm-path latency ≤ 30 ms.

The ONNX export additionally confirms the 577,584 trainable parameter claim by making adapter and head weights inspectable in the ONNX graph.

4) *Hugging Face Open-Science Release (Part 2 Roadmap):* To replace the inaccessible private GCP bucket with a community-accessible distribution:

- **Unified dataset:** Publish AutoevolveAI/Laya-Curriculum-78k as a HuggingFace Dataset with three config subsets (`stage1_security`, `stage2_efficiency`, `stage3_alignment`) and

standardised schema: `{"input_code": str, "target_noul": int, "target_choice": int, "target_score": float}`.

- **LoRA adapter:** Publish standalone PEFT adapter weights at AutoevolveAI/Laya-LoRA-ModernBERT-r8, proving the 577,584 parameter claim directly on the Hub without requiring backbone download.
- **Custom architecture:** Publish `modeling_laya.py` with `trust_remote_code=True` so users load the three-headed model natively via `AutoModel.from_pretrained("AutoevolveAI/Laya-LoRA-ModernBERT-r8", trust_remote_code=True)`.
- **Gradio Space:** Build AutoevolveAI/Laya-System1-Demo — a split-pane UI where the user pastes code; the left pane shows Laya's System 1 decision (p_{noul} , routed agent, energy E) and the right pane conditionally invokes Qwen2.5-Coder-7B *only if* Laya routes the snippet to PASS, demonstrating the energy savings from pre-filtering in real time.

5) *Zenodo Archival (Part 3 Roadmap):* To make SHA-256 provenance claims publicly auditable:

- **Telemetry bundle:** Upload `download_receipt.json`, `training_summary.json`, and `coding_benchmark_and_comparison.json` as a single ZIP to Zenodo. Replace GCS paths in the Data Availability section with the generated DOI (e.g., `10.5281/zenodo.XXXXXXX`).
- **Lean 4 formal proofs:** Upload `formal/ANSE/LayaDecision.lean` and the full `lakefile.lean` to Zenodo. This provides an immutable, version-controlled record of exactly which proof state (including the sorry obligation) was used, enabling independent lake build verification by formal methods researchers.
- **ONNX weights:** Publish the ONNX INT8 model as a Zenodo dataset with its own DOI, separately from the HuggingFace adapter weights, to ensure research longevity independent of HuggingFace platform availability.

6) *Discharging the Lean 4 sorry (Part 4 Roadmap):* To complete the compile-time guarantee for Invariant I3 (energy cost ordering):

- 1) After ONNX export, measure actual inference energy via Intel RAPL or CodeCarbon on the target Cloud Run CPU (e.g., 2 vCPU Intel Cascade Lake, TDP 60 W).

- 2) Write a Lean 4 macro `readJsonEnergy` that reads the Zenodo-published telemetry JSON at lake build time and extracts the scalar `measured_cpu_energy_wh_per_1k` value.
- 3) Inject the concrete value as a let binding and auto-discharge `cpu_energy_bounded` via `norm_num`: example : `cpu_e < qwen7b_e := by norm_num`.
- 4) This transforms Invariant I3 from a structural claim with `sorry` into a machine-verified numerical fact, closing the gap between empirical telemetry and formal MLOps compliance.

X. Conclusion

We have presented Laya-LoRA Coding Companion, a non-autoregressive System 1 decision model trained via three-stage curriculum learning across 10 curated coding datasets (78,503 records). With only 577,584 LoRA-adapted parameters, Laya achieves 91.7% binary gate accuracy on the code quality/security stage, consumes 0.38 Wh per 1,000 queries ($101\times$ less than Qwen2.5-Coder-7B), and operates at \$0.00 idle cost via scale-to-zero Cloud Run deployment.

The GCP Socrate AI data lake (`gs://socrate-ai-datalake/`) provides traceable, SHA-256-verified storage for all datasets, training checkpoints, and inference artefacts. Lean 4 deployment invariants provide compile-time verification of LoRA budget compliance and energy cost ordering before any cloud resource is initialised.

While the 33.3% 12-case benchmark gate accuracy reflects prototype training constraints, the correct blocking of all 4 high-energy code patterns (stubs, SQL/command injection) demonstrates that the architectural decision to use a NAR encoder as a System 1 pre-filter is sound. Full-scale Vertex AI training is the clear next step to realise the operational accuracy targets.

Data Availability

All datasets, training receipts (SHA-256 verified), and model checkpoints are stored in the Socrate AI GCP data lake (`gs://socrate-ai-datalake/`, private; reviewer access available upon request at `xavkal@autoevolveai.dev`). The three provenance JSON receipts are additionally published as a GitHub release artifact at the AutoevolveAI repository (`github.com/xavkal/AutoevolveAI`, tag `laya-coding-companion-v1.0`) to enable independent SHA-256 verification:

- `download_receipt.json` SHA-256: 91662a050d4a501ce2ad82fd5bbdf047922d1463bb3e54f1409796a7b026faf

- `training_summary.json` SHA-256: 00333acc7ea5711c0040cd678dd5704aabc6596b7cb0775
- `coding_benchmark_and_comparison.json` SHA-256: ff546521cef3565203c08470d2aa765f3ac683

Energy Baseline Sources. Qwen2.5-Coder-7B energy (38.5 Wh/1k queries) is derived from Qwen2 technical report [18] GPU inference throughput measurements on NVIDIA A100 hardware. GPT-4o/Claude 3.5 energy (120 Wh/1k) is from the Luccioni et al. carbon footprint study [19]. Both figures are directional estimates for incommensurable deployment configurations (see §5.1 limitations); they are not re-measured in this work.

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